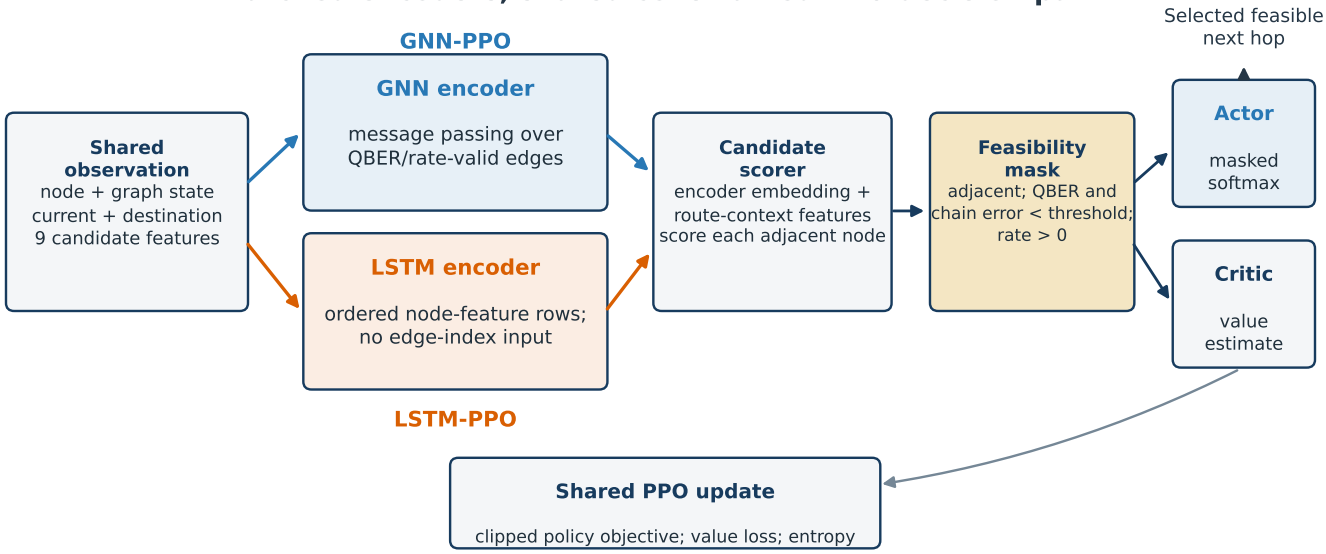
Matched encoders, shared constrained PPO decision path
GNN-PPO
GNN encoder
message passing over QBER/rate-valid edges
Shared observation
node + graph state
current + destination
9 candidate features
LSTM encoder
ordered node-feature rows; no edge-index input
LSTM-PPO
Candidate scorer
encoder embedding + route-context features
score each adjacent node
Feasibility mask
adjacent; QBER and chain error < threshold; rate > 0
Actor
masked softmax
Selected feasible next hop
Critic
value estimate
Shared PPO update
clipped policy objective; value loss; entropy